

JUNQI LU

[GitHub](#) — [Google Scholar](#) — [Personal Blog](#)

Date of Birth: August 24, 2002 — Email: Junqi_Lu@bit.edu.cn

Beijing, China

EDUCATION

Beijing Institute of Technology

Beijing, China

B.E. in Computer Science (Second Bachelor's Degree)

Sep. 2024 – Jun. 2026 (Expected)

- **Ranking:** Ranked 1st in the program.

B.S. in Mathematics and Applied Mathematics

Oct. 2020 – Jun. 2024

- **Thesis Grade:** Excellent (Top tier of graduation projects).
- **Focus:** Rigorous training in Real Analysis, Algebra, and Probability Theory.

PUBLICATIONS

- [1] **Junqi Lu**, Bosen Liu, Cuicui Pei, Qingan Qiu*, and Li Yang*. Learning to optimize termination decisions under hybrid uncertainty of system lifetime and task duration. *Computers & Industrial Engineering*, 2025. DOI: [10.1016/j.cie.2025.111208](https://doi.org/10.1016/j.cie.2025.111208) (Published, IF=6.5, JCR Q1)
- [2] **Junqi Lu**, Ruixiang Sun, Xin Li*, et al. Escaping the Variance Trap: Bisimulation Saddle-Point Optimization for Robust Behavioral Representation Learning. (Under Review)
- [3] **Junqi Lu**, Qingan Qiu*, Rongchi Sun, and Yike Zhang. Simplicity Wins: How Myopic Decisions Enable Safe Learning in Mission-Critical Systems. (Under Review)
- [4] Rongchi Sun, Yike Zhang, **Junqi Lu**, and Qingan Qiu*. Dynamic Switching Strategy for Standby Systems with Switching Time Redundancy. (Under Review)

RESEARCH EXPERIENCE

Deep Reinforcement Learning & State Representation

Jul. 2025 – Present

Research Assistant | Advisor: [Prof. Xin Li](#) | [1029 DRL Lab](#) | Beijing Institute of Technology

- **Theory & Methodology:** Identified the metric learning "Variance Trap" via **SDEs** and proposed **BSPO** (*Bisimulation Saddle-Point Optimization*), a primal-dual framework utilizing an auxiliary network to decouple structural error from transition noise.
- **Evaluation:** Achieved SOTA on **DeepMind Control Suite** under high-noise and visual distractions, significantly outperforming baselines (MICO, SimSR).
- **Outcome:** First-author paper currently under review.

Reliability Engineering & Stochastic Control

Jun. 2023 – Present

Research Assistant | Advisor: [Prof. Qingan Qiu](#) | Beijing Institute of Technology

- **Adaptive Control Framework:** Formulated condition-based task termination for multi-state systems under dual uncertainty as an infinite-horizon **POMDP**, integrating real-time Bayesian inference with right-censored data.
- **Theory & Algorithm:** Proved suboptimality of static policies and derived a closed-form $\mathcal{O}(1)$ **Myopic Heuristic**. This computation-free strategy natively embeds an "Exploration Buffer" and acts as a dynamic risk filter, asymptotically outperforming the complete-information Oracle.
- **Publications:** 1 first-author paper in **CAIE** (Q1, IF=6.5); 1 first-author paper under review; 1 co-authored paper under review.

SELECTED PROJECTS

- Nitrogen Cycling Dynamics Model (MCM-ICM 2025 Finalist)

[GitHub: 100+ Stars]
- **Modeling:** Developed a dynamic ecosystem model using modified **Lotka-Volterra equations** to simulate nitrogen flow across forest and agricultural food webs.
 - **Open Science:** Released a fully reproducible solution repository with comprehensive documentation and tutorials. Garnered **100+ GitHub Stars**, widely recognized by the modeling community for its code quality and educational value.
 - **Award:** Selected as a **Finalist Winner (Top 2%)** out of 27,456 teams globally.

- High-Performance Concurrent BBS & Chat System (C++)

[GitHub Code]
- **Architecture:** Engineered a high-performance, event-driven server on Linux using **Epoll** (I/O Multiplexing) to efficiently manage massive concurrent connections with minimal resource overhead.
 - **Protocol Design:** Designed a custom **Binary Application-Layer Protocol** (Header + Payload) to ensure reliable TCP message framing and implemented a secure handshake mechanism for file transmission.
 - **System Engineering:** Adhered to strict modular design (User/Chat/File Managers) and implemented data persistence logic; realized distinct features including group broadcasting and peer-to-peer file relay.

HONORS & AWARDS

- **Outstanding Student Award** (Top 10% university-wide), BIT

Nov. 2025
- **Academic Excellence Scholarship** (Rank 1st in CS Program), BIT

Oct. 2025
- **Finalist (Top 2%)**, Interdisciplinary Contest in Modeling (ICM)

Feb. 2025
- **University "Excellent Thesis" Prize**, BIT

Jun. 2024
- **Meritorious Winner (Top 9%)**, Mathematical Contest in Modeling (MCM)

Feb. 2023
- **Second Prize**, China Undergraduate Mathematical Contest in Modeling (CUMCM)

Nov. 2022

RELEVANT COURSEWORK

- **Computer Science:** Data Structures and Algorithms, Operating Systems, Computer Architecture, Computer Networks, Object-Oriented Programming, Database Principles.
- **Mathematics:** Real Analysis, Functional Analysis, Abstract Algebra, Probability Theory, Mathematical Statistics, Partial Differential Equations (PDE), General Topology, Numerical Analysis.
- **Specialized Topics:** Reinforcement Learning (Graduate-Level), Machine Learning Fundamentals.

SKILLS

Programming Languages	Python (Expert in PyTorch, NumPy, Pandas), C++ (Linux System Dev), SQL, MATLAB, Bash
AI Workflow	Chinese (Native), English (B2), German (A1)
Tools	Proficient in AI-assisted coding (VS Code + Copilot/Cursor) for efficient development.
Outreach	Git, Docker, LaTeX, Makefiles, Lean 4 (Formal Proof Verification)
	Tech Blogger (CNBlogs) with 20K+ views (20+ articles on DSA & LeetCode).

INTERESTS

- **Marathon Running:** Committed long-distance runner with an annual mileage exceeding **1,000 km** for three consecutive years. Half-Marathon Personal Best: **1:41:34**.
- **Music:** Lead guitarist and bassist for university rock bands. Passionate about classic rock arrangement and performance; active participant in campus music festivals.